

Homework Assignment 2 (2/20/2009)

25 Points — Due 3/4/2009 (by 1pm)

- (i) (10 Points) Similar to Section 2.2 in our Lecture Notes, provide a brief (at most one page each) summary of a historical contributor to statistical graphics and of a currently active researcher in statistical graphics. Include a painting or a photo, an important plot, and summarize the main contributions of this person. For a current researcher, you should include a link to this person's personal Web page.

Use Michael Friendly's Milestones Web page at <http://www.math.yorku.ca/SCS/Gallery/milestone/milestone.pdf> to identify suitable candidates for this summary.

Each homework group should deal with different persons than the other groups. You have to coordinate by 2/27/2009 at the latest which group is going to work with which historical and current persons. If two (or more) groups want to provide a summary of the same person, flip a coin!

- (ii) (10 Points) Revisit our R code from Section 3.1 from our Lecture Notes at http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/Ch3_RGBColors.R. Choose 10 color schemes (of 8 colors each) as follows:

- 2 schemes related to the pure R, G, and B schemes used in class (you only need 8 colors; we used 10 colors in class)
- 2 sequential schemes from RcolorBrewer
- 2 diverging schemes from RcolorBrewer
- 2 sequential schemes from colorspace
- 2 diverging schemes from colorspace

Arrange your 10 plots of the same data, but using these different color schemes, in a 5 rows $\times$ 2 column layout on a single page. Clearly label which color scheme has been used for each plot, similar to the ColorBrewer names used in Section 3.4.1 of our Lecture Notes. Write the output into a pdf file via the `pdf` command in R.

Now, consider the following 4 viewing options:

- print the pdf on a color printer
- print the pdf on a black and white printer (or copy the color printout on a black and white copy machine)
- look at the pdf on a PC
- look at the pdf on a laptop

For each of these four different viewing options, individually rank the color visibility. Which plot best shows 8 different colors (1), which is second best (2), ..., which is worst (10). Ties are not allowed.

This ranking should be done by all group members under the same conditions, e.g., at the same computer under the same lighting conditions. Ideally, all group members should simultaneously look at the 10 plots of a particular viewing option and then rank the 10 plots independently.

First tabulate your individual rankings for each of the 10 plots for each of the 4 different viewing options for each of the 2 (or 3) students. Then calculate average ranks for each viewing option and finally calculate the overall average rank over all 4 viewing options. Include these tables as part of your homework submission.

Which color scheme is the best under the 4 viewing options? And which is the best across all 4 viewing options? Is your overall winner also a winner under 1 (or more) of the 4 viewing options, or is the overall winner always highly ranked, but not necessarily an individual winner?

- (iii) (5 Points) As this is a Graphics class, also provide a graphical representation of your rankings. This figure should contain rankings for the 10 color scheme choices $\times$ the 4 viewing options $\times$ the 2 (or 3) students. Also display the average ranks for each viewing option and the overall average rank.

General Submission Rules: (for homework, EC, and individual reports)

Please follow the general submission rules from Homework 1. In addition, please follow these instructions:

- Make sure to include all graphical results into your pdf submission. Link to the R files, assuming they are in the same directory as the main tex file.
- Include a title page, including course name, student names, number of homework, and the submission date.
- Start answers to each question on a new page, clearly labeling which question is being answered.
- Define a toggle (say `tasksbystudent`) that allows you to switch on/off the following information:

On the very last page of your submission, provide a table that summarizes for each question how much each group member has contributed to each of the main tasks of that question, e.g., Question x: R coding (Student A: 80%, Student B: 20%), $\text{\LaTeX}$ typesetting (A: 10%, B: 90%), Other tasks, such as selection of color schemes (A: 50%, B: 50%).

It is totally up to you how to split tasks, even up to the extreme case that one student does 100% of all tasks for one question and, thus, another student does 100% of all tasks for the next question. However, each student should do more than 50% of the work for each of the main tasks (in particular R coding and $\text{\LaTeX}$ typesetting) at some point. So, a submission where one student does 100% (or just 51%) of all R coding for all questions on an assignment while another student does 100% (or just 51%) of all $\text{\LaTeX}$ typesetting for all questions is *not* acceptable.

Thus, decide in advance who will take the lead for each of the main tasks on a particular assignment sheet. If I nevertheless see such an uneven distribution of tasks, I will ask the trailing student(s) to work on some additional questions to become more familiar with a particular main task of the homework assignment.